1. Profit Calculation
   a. Create a numeric vector named original_price with the given values:
      9, 24, 245, 345, 56, 45, 100, 86, 59,20
   b. Create a new vector named selling_price by adding a 15% profit to each item in the original_price vector.
   c. Create a numeric Vector named quantity_sold representing the quantity of each item sold:
      1,5,8,4,3,2,5,4,8,5
   d. Create a new vector named profit by multiplying the 15% profit margin by the original_price and then multiplying by the quantity_sold.
   e. Use the cbind function to combine the original original_price, selling_price, quantity_sold, and profit into a matrix. Display the matrix to see the side-by-side comparison.
   f. Use the paste function to display the total profit for the day.

```
      original_price selling_price quantity_sold profit
 [1,]              9         10.35             1   1.35
 [2,]             24         27.60             5  18.00
 [3,]            245        281.75             8 294.00
 [4,]            345        396.75             4 207.00
 [5,]             56         64.40             3  25.20
 [6,]             45         51.75             2  13.50
 [7,]            100        115.00             5  75.00
 [8,]             86         98.90             4  51.60
 [9,]             59         67.85             8  70.80
[10,]             20         23.00             5  15.00

"The profit for this day is  771.45"
```

2. Daily Financial Analysis
   a. Create a numeric vector named Day representing the days of the week: Day 1,2,3,4,5
   b. Create a numeric vector named daily_money with a constant value of 200 for each day.
   c. Create a numeric vector named daily_expenses representing daily expenses: 150, 120, 160, 142, 98
   d. Create a vector named daily_savings by subtracting daily_expenses from daily_money.
   e. Create a vector named daily_savings_percentage by calculating the percentage of savings relative to daily money.
   f. Use cbind to combine Day, daily_money, daily_expenses, daily_savings, and daily_savings_percentage into a matrix.
   g. Calculate the total savings for the week by summing the daily_savings vector.
   h. Display the total savings using the paste function.

```
     Day daily_money daily_expenses daily_savings daily_savings_percentage
[1,]   1         200            150            50                       25
[2,]   2         200            120            80                       40
[3,]   3         200            160            40                       20
[4,]   4         200            142            58                       29
[5,]   5         200             98           102                       51

[1] "Total Savings for the week:  330"
```

3. Monthly Energy Cost Calculation
   a. Create a numeric vector named Month representing the months of the year:
      1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12
   b. Create a numeric vector named energy_consumed representing the energy consumption for each month: 150, 200, 145, 210, 145, 178, 136, 145, 120, 189, 136, 168
   c. Set a constant value for the cost per kWh: 12 pesos

d. Use the formula monthly_energy_cost = energy_consumed * cost_per_kWh to calculate the monthly energy cost.
e. Use cbind to combine Month, energy_consumed, cost_per_kWh, and monthly_energy_cost into a matrix.

	Month	energy_consumed	cost_per_kWh	monthly_energy_cost
[1,]	1	150	12	1800
[2,]	2	200	12	2400
[3,]	3	145	12	1740
[4,]	4	210	12	2520
[5,]	5	145	12	1740
[6,]	6	178	12	2136
[7,]	7	136	12	1632
[8,]	8	145	12	1740
[9,]	9	120	12	1440
[10,]	10	189	12	2268
[11,]	11	136	12	1632
[12,]	12	168	12	2016

4. Loan Payment Calculation
   a. Create a sequence of loan amounts ranging from 30,000 to 100,000 in increments of 1,000.
   b. Calculate the fixed 4% interest for each loan amount.
   c. Calculate monthly payments for 1, 2, 3, 4, and 5-year loan terms using the fixed 4% interest.
   d. Use cbind to combine Loaned Amount, Fixed 4% Interest, and monthly payments for different terms into a matrix.

	Loaned_Amount	Fixed_4_percent_interest	For_1_year_Term	For_2_year_Term	For_3_year_Term	For_4_year_Term	For_5_year_Term
[1,]	30000	1200	2600.00	1350	933.33	725.00	600
[2,]	31000	1240	2686.67	1395	964.44	749.17	620
[3,]	32000	1280	2773.33	1440	995.56	773.33	640
[4,]	33000	1320	2860.00	1485	1026.67	797.50	660
[5,]	34000	1360	2946.67	1530	1057.78	821.67	680
[6,]	35000	1400	3033.33	1575	1088.89	845.83	700
[7,]	36000	1440	3120.00	1620	1120.00	870.00	720
[8,]	37000	1480	3206.67	1665	1151.11	894.17	740
[9,]	38000	1520	3293.33	1710	1182.22	918.33	760
[10,]	39000	1560	3380.00	1755	1213.33	942.50	780
[11,]	40000	1600	3466.67	1800	1244.44	966.67	800
[12,]	41000	1640	3553.33	1845	1275.56	990.83	820
[13,]	42000	1680	3640.00	1890	1306.67	1015.00	840
[14,]	43000	1720	3726.67	1935	1337.78	1039.17	860
[15,]	44000	1760	3813.33	1980	1368.89	1063.33	880
[16,]	45000	1800	3900.00	2025	1400.00	1087.50	900
[17,]	46000	1840	3986.67	2070	1431.11	1111.67	920
[18,]	47000	1880	4073.33	2115	1462.22	1135.83	940
[19,]	48000	1920	4160.00	2160	1493.33	1160.00	960
[20,]	49000	1960	4246.67	2205	1524.44	1184.17	980
[21,]	50000	2000	4333.33	2250	1555.56	1208.33	1000
[22,]	51000	2040	4420.00	2295	1586.67	1232.50	1020
[23,]	52000	2080	4506.67	2340	1617.78	1256.67	1040
[24,]	53000	2120	4593.33	2385	1648.89	1280.83	1060
[25,]	54000	2160	4680.00	2430	1680.00	1305.00	1080
[26,]	55000	2200	4766.67	2475	1711.11	1329.17	1100
[27,]	56000	2240	4853.33	2520	1742.22	1353.33	1120
[28,]	57000	2280	4940.00	2565	1773.33	1377.50	1140
[29,]	58000	2320	5026.67	2610	1804.44	1401.67	1160
[30,]	59000	2360	5113.33	2655	1835.56	1425.83	1180
[31,]	60000	2400	5200.00	2700	1866.67	1450.00	1200
[32,]	61000	2440	5286.67	2745	1897.78	1474.17	1220
[33,]	62000	2480	5373.33	2790	1928.89	1498.33	1240
[34,]	63000	2520	5460.00	2835	1960.00	1522.50	1260
[35,]	64000	2560	5546.67	2880	1991.11	1546.67	1280
[36,]	65000	2600	5633.33	2925	2022.22	1570.83	1300
[37,]	66000	2640	5720.00	2970	2053.33	1595.00	1320
[38,]	67000	2680	5806.67	3015	2084.44	1619.17	1340
[39,]	68000	2720	5893.33	3060	2115.56	1643.33	1360
[40,]	69000	2760	5980.00	3105	2146.67	1667.50	1380
[41,]	70000	2800	6066.67	3150	2177.78	1691.67	1400
[42,]	71000	2840	6153.33	3195	2208.89	1715.83	1420
[43,]	72000	2880	6240.00	3240	2240.00	1740.00	1440
[44,]	73000	2920	6326.67	3285	2271.11	1764.17	1460
[45,]	74000	2960	6413.33	3330	2302.22	1788.33	1480
[46,]	75000	3000	6500.00	3375	2333.33	1812.50	1500
[47,]	76000	3040	6586.67	3420	2364.44	1836.67	1520
[48,]	77000	3080	6673.33	3465	2395.56	1860.83	1540
[49,]	78000	3120	6760.00	3510	2426.67	1885.00	1560
[50,]	79000	3160	6846.67	3555	2457.78	1909.17	1580
[51,]	80000	3200	6933.33	3600	2488.89	1933.33	1600
[52,]	81000	3240	7020.00	3645	2520.00	1957.50	1620
[53,]	82000	3280	7106.67	3690	2551.11	1981.67	1640
[54,]	83000	3320	7193.33	3735	2582.22	2005.83	1660
[55,]	84000	3360	7280.00	3780	2613.33	2030.00	1680
[56,]	85000	3400	7366.67	3825	2644.44	2054.17	1700
[57,]	86000	3440	7453.33	3870	2675.56	2078.33	1720
[58,]	87000	3480	7540.00	3915	2706.67	2102.50	1740
[59,]	88000	3520	7626.67	3960	2737.78	2126.67	1760
[60,]	89000	3560	7713.33	4005	2768.89	2150.83	1780
[61,]	90000	3600	7800.00	4050	2800.00	2175.00	1800
[62,]	91000	3640	7886.67	4095	2831.11	2199.17	1820
[63,]	92000	3680	7973.33	4140	2862.22	2223.33	1840
[64,]	93000	3720	8060.00	4185	2893.33	2247.50	1860
[65,]	94000	3760	8146.67	4230	2924.44	2271.67	1880
[66,]	95000	3800	8233.33	4275	2955.56	2295.83	1900
[67,]	96000	3840	8320.00	4320	2986.67	2320.00	1920
[68,]	97000	3880	8406.67	4365	3017.78	2344.17	1940
[69,]	98000	3920	8493.33	4410	3048.89	2368.33	1960
[70,]	99000	3960	8580.00	4455	3080.00	2392.50	1980
[71,]	100000	4000	8666.67	4500	3111.11	2416.67	2000

```
original_price<- c(9,24,245,345,56,45,100,86,59,20)

selling_price<-print(original_price+(original price*0.15))

quantity_sold<-c(1,5,8,4,3,2,5,4,8,5)

profit<-print((original_price*0.15)*quantity_sold)

cbind(original_price,selling_price,quantity sold, profit)

total_profit<-print(sum(profit))

paste("The profit for this day is", total_profit)
```

1

```
Day<-c(1,2,3,4,5)

daily_money<-print(200)

daily_expenses<-c(150,120,160,142,98)

daily_savings<-print(dally_money-daily_expenses)

daily_savings_percentage<-((daily_savings/daily_money)*100)

cbind(Day, daily_money, daily_expenses, daily_savings, daily_savings_percentage)

total_savings<-(sum(daily_savings_percentage))

paste("total savings for the week:", total_savings)
```

2

```
Months<-c(1,2,3,4,5,6,7,8,9,10,11,12)

energy_consumed<-c(150,200,145,210,145,178,136,145,120,189,136,168)

cost_per_kwh<-(12)

monthly_energy_cost<-(energy_consumed*12)

cbind(Month, energy_consumed, cost_per_kwh, monthly_energy_cost)
```

3

```
loan_amounts<-(seq(30000, 100000, 1000))

fixed_interest<-print(loan_amounts*.04)

for_1_year_term<-print((fixed_interest+loan_amounts)/12)

for_2_year_term<-print(((fixed_interest*2)+loan_amounts)/24)

for_3_year_term<-print(((fixed_interest*3)+loan_amounts)/36)

for_4_year_term<-print(((fixed_interest*4)+loan_amounts)/48)

for_5_year_term<-print(((fixed_interest*5)+loan_amounts)/60)

cbind(loan_amounts,fixed_interest,for_1_year_term, for_2_year_term, for_3_year_term,
for_4_year_term, for_5_year_term)
```

4